MASTER'S THESIS

Environment-Aware Conditioned Random Walks for Animal-Movement Analysis

A modular pipeline from environmental annotation to utilization distributions and visit probabilities

Omar Chatila

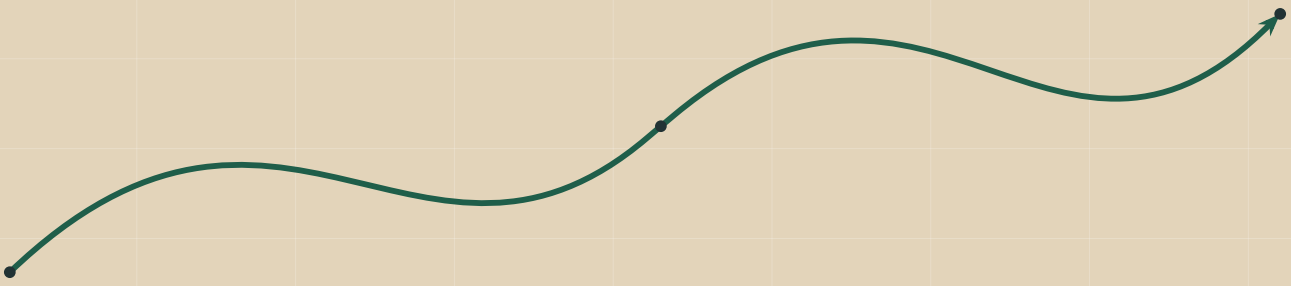
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Declaration of authorship

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Publications and author contributions

Parts of this thesis build on the coauthored manuscript

> Kevin Buchin, Mart Hagedoorn, and Omar Chatila. "Advanced Discretized Conditional Random Walks for Utilization Distributions, Visit Probabilities, and Context-Aware Transitions". Unpublished manuscript, 2026.

The manuscript serves as a theory and notation reference for the conditioned-random-walk core, including environment-aware transitions. The thesis is broader and application-led: it develops the complete preprocessing pipeline, MoveApps integration, ecological case studies, and additional parallel and GPU algorithms for spatial-summary computation (Buchin, Hagedoorn, and Chatila 2026).

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Writing prompt

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Abstract

Animal tracking data are often sparse or irregular, leaving substantial uncertainty about movement between observed locations. This thesis develops and evaluates a modular pipeline for reconstructing plausible movement with environment-aware, endpoint-conditioned random walks. The pipeline combines environmental annotation, trajectory segmentation, behavioral-state inference, transition-kernel construction, random-walk interpolation, and the computation of utilization distributions and visit probabilities. It is implemented as reusable components for the MoveApps ecosystem and evaluated with animal-movement case studies and computational benchmarks.

Replace this provisional abstract last. State the problem, method, data, principal quantitative results, and ecological/computational implications in approximately 200–300 words. Avoid citations and undefined abbreviations.

Keywords. animal movement; movement ecology; random walks; trajectory interpolation; environmental annotation; utilization distribution; MoveApps

Acknowledgements

Thank supervisors, collaborators, data providers, infrastructure providers, and funding bodies. Distinguish scientific contribution, technical support, and personal support. Check the required wording for funded work and externally supplied data.

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Acronyms and abbreviations

API application programming interface

BBMM Brownian bridge movement model

CPU central processing unit

CRS coordinate reference system

DP dynamic programming

ESA European Space Agency

GPS Global Positioning System

GPU graphics processing unit

HMM hidden Markov model

KDE kernel density estimate

MCMC Markov chain Monte Carlo

STAC SpatioTemporal Asset Catalog

UD utilization distribution

UTM Universal Transverse Mercator

1 Introduction

1.1 Motivation

Modern animal-borne sensors can record movement at unprecedented spatial and temporal resolution, yet real trajectories remain incomplete. Battery constraints, failed fixes, transmission loss, environmental obstruction, and deliberately sparse sampling create gaps between known locations. A straight segment closes such a gap geometrically but suppresses uncertainty and ignores the animal's movement capacity, behavioral state, and surrounding landscape. Ecological analyses based on reconstructed paths therefore require a model that represents many plausible movements rather than one visually convenient line.

Endpoint-conditioned movement models address this need by describing possible paths between two observations at fixed times. Classical Brownian bridge movement models (BBMMs) provide a principled probabilistic construction, but more flexible transition rules are needed when movement depends on direction, behavioral state, terrain, barriers, or time-varying environmental covariates. Discretized conditional random walks offer such flexibility by combining transition kernels with dynamic programming (DP) (Buchin, Hagedoorn, and Korn 2024).

This thesis considers the full analytical chain required to turn tracking records into ecologically interpretable spatial summaries. It connects environmental annotation, segmentation and behavioral inference, movement-kernel estimation, conditioned random-walk interpolation, and the derivation of utilization distributions (UDs) and visit probabilities. The components are designed for independent use in MoveApps workflows as well as for end-to-end analyses.

1.2 Problem statement

Let an observed trajectory be a time-ordered sequence

$$\mathcal{T} = \big((\mathbf{x}_0, t_0), \ldots, (\mathbf{x}_n, t_n)\big), \qquad \mathbf{x}_i \in \mathbb{R}^2, \quad t_i < t_{i+1}. \tag{1.1}$$

Each interval may additionally be associated with environmental covariates, inferred behavioral states, and uncertainty. The central task is to construct a distribution over discrete paths that (i) start and end at the observed locations, (ii) respect the elapsed time, and (iii)

follow locally appropriate transition rules. Those path distributions must then support sampled interpolations, time-integrated space-use estimates, and event probabilities.

Writing prompt

State the formal input, output, conditioning assumptions, and scientific estimand in one place. Be explicit about whether the method estimates plausible paths, expected occupancy, long-term home range, or a finite-interval space-use distribution; these are not interchangeable.

1.3 Research objectives and questions

The overarching objective is to design, implement, and evaluate an environment-aware conditioned-random-walk pipeline suitable for reproducible animal-movement analysis. The thesis is organized around the following provisional research questions.

RQ1: How can heterogeneous tracking and environmental data be transformed into a consistent spatial and temporal representation without discarding ecologically relevant context?

RQ2: Which segmentation and behavioral-annotation strategies yield stable, interpretable inputs for state- or terrain-dependent transition kernels?

RQ3: How should empirical movement be converted into discrete kernels while preserving physical scale, directional persistence, and uncertainty?

RQ4: How can endpoint-conditioned random walks incorporate spatial barriers and context-dependent transitions efficiently?

RQ5: How do the resulting interpolations, UDs, and visit probabilities compare with established baselines in predictive quality, calibration, robustness, and runtime?

Writing prompt

Refine these questions after fixing the final experiments. Every question should map to a method, an evaluation metric, and a result subsection.

1.4 Contributions

Replace this list with the final, evidence-backed contributions. Separate algorithmic, software-engineering, empirical, and ecological contributions. Use verbs such as “introduces”, “proves”, “implements”, or “evaluates” only when the corresponding evidence appears later.

- A modular preprocessing workflow for environmental and behavioral annotation of animal trajectories.
- Data-driven construction of state- and context-dependent movement kernels.
- Endpoint-conditioned random-walk methods for interpolation, utilization distributions, and visit probabilities.

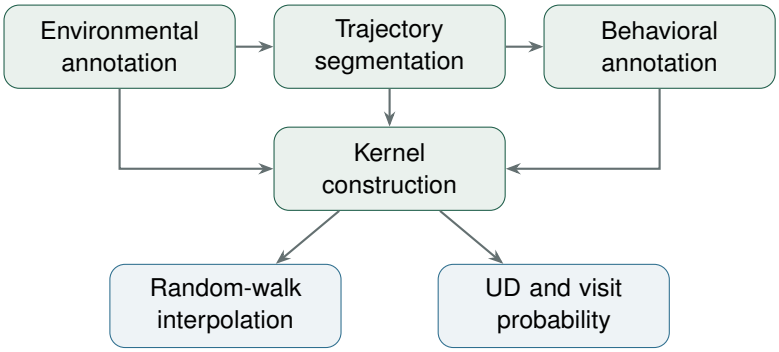


Figure 1.1: Conceptual pipeline of the thesis. Components may be composed end-to-end or reused independently in a MoveApps workflow.

- Parallel CPU and GPU algorithms for utilization-distribution computation, including a memory-bounded aggregate.
- A MoveApps-oriented software architecture with reproducible benchmark and case-study workflows.

1.5 Scope and terminology

The primary application domain is animal movement, although the computational framework is not species-specific. The thesis distinguishes an observed *fix* from an unobserved intermediate position, a sampled *walk* from its probability distribution, and a finite-interval occupancy distribution from an asymptotic home-range estimate. Environmental *barriers* may be hard (transitions forbidden) or soft (transitions down-weighted). A *kernel* denotes a normalized discrete distribution over possible displacements, not a smoothing kernel unless stated otherwise.

1.6 Thesis structure

Chapter 2 establishes the ecological and algorithmic background. Chapters 3 to 5 describe the three preprocessing and estimation stages. Chapters 6 and 7 present the conditioned-walk computations and derived spatial summaries. Chapter 8 documents integration and reproducibility, while Chapters 9 and 10 define and report the evaluation. Chapters 11 and 12 interpret the findings and conclude the thesis.

2 Background and Related Work

2.1 Animal movement as a sampled process

Animal-movement data are discrete observations of an underlying continuous process. Their interpretation depends on the sampling schedule, positional error, spatial scale, and the relation between the observation interval and the behavior of interest. Missingness may also be informative: canopy cover, topography, diving, or device orientation can affect fix success. A movement model should therefore state which source of uncertainty it represents and which remain outside its scope.

Writing prompt

Introduce the focal ecological questions and data regimes. Include a concise table of species or case studies, fix interval, spatial extent, sample size, environmental layers, and intended role in the evaluation.

2.2 Environmental context and habitat selection

Habitat covariates can be attached to observed fixes, possible endpoints, complete steps, or every intermediate model state. These choices imply different ecological questions. Step-selection functions compare observed movement steps with available alternatives and link local movement decisions to environmental covariates (Thurfjell et al. 2014). Integrated approaches jointly estimate aspects of movement and habitat selection, while state-dependent approaches allow preferences to vary with latent behavior (Pohle et al. 2024).

In this thesis, environmental annotation serves two related but distinct roles. First, it provides descriptive covariates for segmentation and behavioral inference. Second, it may directly alter the transition probabilities of the random walk. The second role requires particular care because annotation error or arbitrary raster resolution propagates into the modeled path distribution.

2.3 Segmentation and behavioral states

Trajectory segmentation partitions a track into internally coherent intervals. Criteria-based algorithms can express constraints involving duration, speed, heading, or spatial range

(Alewijnse et al. 2014). Exploratory approaches such as behavioral change-point analysis and fitted multi-state movement models make different assumptions and produce different kinds of boundaries; comparing their output is therefore more informative than treating one method as ground truth (Gurarie et al. 2016).

Hidden Markov models (HMMs) represent behavior as a latent Markov chain whose states emit observed movement features. State labels such as "resting", "foraging", and "traveling" require ecological interpretation after fitting: numerical state identifiers have no intrinsic meaning. The selected number of states, observation distributions, initialization, and treatment of irregular time intervals can materially alter downstream kernels.

2.4 Movement bridges and random walks

A BBMM conditions Brownian motion on two observed locations and their times. It yields a probability distribution for intermediate locations and has been widely used for space-use estimation (Horne et al. 2007). Dynamic variants allow the motion variance to vary along the track (Kranstauber et al. 2012). These models offer a strong baseline but do not directly provide arbitrary location-, time-, direction-, state-, and terrain-dependent transition matrices.

The discretized framework of Buchin, Hagedoorn, and Korn (2024) represents movement through finite displacement kernels on a spatial grid. Forward probabilities are computed by DP; a path ending at the required destination can then be sampled backward. This construction supports Brownian, biased, correlated, and mixed walks under a common computational interface.

2.5 Utilization distributions and visit probabilities

A UD is a probability distribution over an animal's location in a defined temporal and spatial domain. The term is used for several estimands, including empirical occupancy, smoothed location density, bridge-based finite-interval use, and stationary space use. These should be separated in the evaluation. Autocorrelated-kernel approaches such as ctmm provide a prominent reference family for home-range estimation (Calabrese et al. 2016).

Visit probability is event-based rather than time-integrated: for a region A and a conditioned path $X_{0:T}$, a typical quantity is

$$P_{\text{visit}}(A) = \Pr(\exists\, t \in \{0, \dots, T\} : X_t \in A \mid X_0 = \mathbf{s}, X_T = \mathbf{e}) \,. \tag{2.1}$$

Repeated occupancy and at-least-once visitation answer different ecological questions and must not be conflated.

2.6 MoveApps and modular scientific workflows

MoveApps provides composable processing components for animal-movement data. Modularization enables an environmental-annotation or segmentation component to be reused independently, makes task-specific configuration visible, and supports alternative workflows. It also creates interface obligations: coordinate systems, time zones, identifiers, missing values, units, and metadata must be preserved consistently between apps.

Writing prompt

Add the authoritative MoveApps citation and describe its data contract, execution model, artifact handling, and versioning once the final app interfaces are fixed.

2.7 Research gap

Synthesize—not merely list—the limitations of existing interpolation and space-use methods. End with the precise gap addressed by the thesis: flexible endpoint conditioning with environmental and behavioral context, delivered as reusable and evaluated workflow components.

3 Data Preparation and Environmental Annotation

3.1 Input data and study systems

The pipeline accepts time-stamped point trajectories grouped by individual. At minimum, each record requires an individual identifier, timestamp, two-dimensional position, and an explicitly declared coordinate reference system (CRS). Optional attributes include taxon, positional uncertainty, existing behavioral labels, and environmental measurements.

3.2 Quality control and normalization

Preprocessing should make all exclusions auditable. The workflow sorts fixes by individual and time, detects duplicate timestamps and invalid geometries, validates the input CRS, records time-zone handling, and reports missing fields. Filters for implausible speed, positional error, or study-specific periods are applied only under explicit rules.

Writing prompt

Specify the exact validation policy. Report the number of records before and after each filter and explain whether dropped fixes can create new apparent gaps.

Table 3.1: Planned study-system inventory. Replace provisional entries with final data and licensing details.

Study / taxon	Fix interval	Spatial extent	Pipeline role	Source / licence
Case study A	—	—	Method development	—
Case study B	—	—	Held-out evaluation	—
Case study C	—	—	Stress test	—

3.3 Spatial reference systems

Geographic coordinates are appropriate for storage and data exchange, whereas distance-based movement models require a projected metric space. For local terrestrial studies, the implementation selects one Universal Transverse Mercator (UTM) zone per individual and applies it consistently to its fixes and rasters. Tracks spanning multiple zones or approaching the poles require an alternative projection or a long-range strategy because a single local UTM projection may introduce unacceptable distortion.

Define projection-selection criteria quantitatively. Include distortion diagnostics or a maximum acceptable extent, and document how studies crossing the antimeridian are handled.

3.4 Land-cover acquisition and raster processing

The local annotation mode retrieves European Space Agency (ESA) WorldCover layers through a SpatioTemporal Asset Catalog (STAC) interface, clips them to an animal-specific area of interest, reprojects them into metric coordinates, and samples the resulting raster at each fix. A long-range mode uses a coarser land polygon to distinguish land and water for wide-ranging, aerial, or marine tracks. The modes intentionally make different accuracy–coverage trade-offs.

The raster-processing sequence is:

1. compute and validate the trajectory extent in WGS 84;
2. obtain or construct the environmental raster for that extent;
3. reproject the raster and trajectory into a common metric CRS;
4. create the discrete analysis grid;
5. sample classes at fixes and retain grid indices and provenance metadata.

3.5 Environmental representation for movement models

The annotation at an observed fix is not sufficient for an environment-aware interpolation. The transition model requires context over all relevant grid cells and, for dynamic covariates, over time. Let

$$\mathbf{z}(x, y, t) = \big(z_1(x, y, t), \ldots, z_p(x, y, t)\big) \tag{3.1}$$

denote the environmental field aligned with the random-walk grid. Categorical layers may determine terrain classes or barriers; continuous layers may be transformed into bias, diffusion, or kernel-mixture parameters.

Writing prompt

For every environmental product, report provider, product version, reference year/time, native resolution, resampling method, no-data handling, and attribution. Separate remotely sensed classification uncertainty from movement-model uncertainty.

3.6 Validation

Annotation validation combines geometric and ecological checks: overlay plots of fixes and rasters, class-frequency summaries, inspection of coastline and boundary cells, round-trip coordinate tests, and known-location spot checks. Tests should include high latitudes, tracks crossing a projection boundary, small islands, and points exactly on raster edges.

Insert a compact validation figure with the original geographic track, projected raster, sampled classes, and discrete grid for one terrestrial and one marine or aerial example.

4 Trajectory Segmentation and Behavioral Annotation

4.1 Why separate gaps, segments, and states?

Temporal gaps, coherent movement segments, and behavioral states are related but distinct. Gap splitting prevents unsupported long steps from entering feature estimation. Segmentation locates intervals of relatively homogeneous movement. Behavioral annotation assigns an interpretable or latent state to fixes or segments. Conflating these operations can make a downstream kernel reflect sampling artifacts instead of movement.

4.2 Gap-based preprocessing

A baseline rule splits an individual's ordered trajectory whenever the elapsed time $\Delta t_i = t_{i+1} - t_i$ exceeds a threshold, for example a multiple of the individual's typical fix interval. This simple rule is useful for data hygiene but cannot detect behavioral changes within regularly sampled data.

Writing prompt

Define the estimator of the typical interval, the threshold multiplier, minimum segment length, and handling of irregular schedules. Justify the choices with a sensitivity analysis rather than relying on a default alone.

4.3 Criteria-based segmentation

Criteria-based segmentation defines acceptable segments through temporal or geometric rules, including minimum duration, spatial containment, speed ranges, and Boolean combinations. Stable-criteria frameworks can support expressive rules while retaining efficient computation (Alewijnse et al. 2014). These rules are useful when domain knowledge yields a defensible behavioral definition.

Table 4.1: Behavioral-model reporting checklist.

Component	Required report
Features	Definitions, units, transforms, scaling, and missing-value policy
States	Candidate counts, selection rule, occupancy, and interpretation
Fitting	Initialization, optimizer, convergence, and random seeds
Temporal model	Transition matrix, dwell-time implications, and sampling assumptions
Validation	Held-out likelihood, stability, diagnostics, and expert comparison if available
Output	State indexing, uncertainty/posteriors, change points, and unclassified records

4.4 Data-driven behavioral models

The implementation supports complementary data-driven approaches:

Hidden Markov model
A latent state sequence with state-dependent observation distributions and Markovian transitions.

Behavioral change-point analysis
Detection of changes in movement parameters, followed by segment clustering.

Hybrid segmentation
Change points determine contiguous segments, while posterior evidence from an HMM assigns a state to each segment.

Candidate movement features include step length, speed, turning angle, directional persistence, angular diffusivity, and terrain class. Features computed from adjacent fixes are affected by missingness and unequal time intervals; their definitions and units must therefore be specified explicitly.

4.5 Model selection and state interpretation

The number of latent states is not merely a numerical tuning parameter. Too few states may combine ecologically distinct modes, while too many can create poorly supported kernels and unstable labels. Selection may combine information criteria, out-of-sample likelihood, parameter stability, state occupancy, dwell-time behavior, and ecological interpretability. State labels are assigned only after examining their feature distributions and representative track segments.

4.6 Output contract

Each annotated record retains the original identifier, timestamp, geometry, and environmental columns, and adds a numeric state, optional human-readable state name, segment identifier, and change-point indicator. Unusable or unclassified records receive an explicit sentinel value rather than being silently removed. Metadata records the method and fitted configuration.

4.7 Evaluation of segmentation and annotation

When expert labels exist, state matching must account for label permutation before computing accuracy, macro-averaged F-scores, or confusion matrices. Without ground truth, evaluation should focus on reproducibility, stability under resampling or initialization, segment-duration plausibility, predictive fit, and sensitivity of downstream kernels and UDs.

Insert the final comparison among gap-only, criteria-based, HMM, change-point, and hybrid methods. Avoid presenting agreement between unsupervised methods as proof of biological truth.

5 Movement-Kernel Construction

5.1 From empirical steps to transition kernels

For consecutive projected locations, let

$$\Delta \mathbf{x}_i = \mathbf{x}_{i+1} - \mathbf{x}_i, \qquad \ell_i = \|\Delta \mathbf{x}_i\|_2, \qquad \theta_i = \operatorname{atan2}(\Delta y_i, \Delta x_i). \tag{5.1}$$

A movement kernel describes a distribution over future displacement, possibly conditioned on the incoming direction, behavioral state, terrain, or time. Training steps must correspond to a known physical interval; changing the model interval by naively multiplying or dividing observed displacements changes the process being estimated.

5.2 Kernel families

5.2.1 Brownian kernels

An isotropic Brownian kernel assigns probability according to displacement magnitude and a diffusion parameter. It provides a low-complexity reference without directional persistence.

5.2.2 Correlated kernels

Correlated random walks express direction relative to the previous heading, allowing persistence or reversal. The discrete implementation partitions incoming headings into direction bins and associates each bin with an appropriately rotated transition distribution.

5.2.3 Mixtures and state dependence

A mixture of K components can capture multimodal step patterns:

$$f(\Delta \mathbf{x} \mid s) = \sum_{k=1}^{K} \pi_{s,k}\, \mathcal{N}(\Delta \mathbf{x}; \boldsymbol{\mu}_{s,k}, \boldsymbol{\Sigma}_{s,k}), \qquad \sum_{k=1}^{K} \pi_{s,k} = 1. \tag{5.2}$$

The conditioning variable s may denote an inferred behavioral state, a terrain class, or another context. A state-specific kernel is estimable only when that state contains enough valid transitions.

5.3 Discretization and physical support

For grid-cell width c and physical kernel radius R, the discrete support radius is

$$S = \left\lceil \frac{R}{c} \right\rceil . \tag{5.3}$$

Rather than fixing one arbitrary window for all kernels, R may be chosen as the smallest centered region containing a specified probability mass. The continuous density is then integrated or sampled onto discrete displacement cells and normalized.

Writing prompt

Document whether rasterization uses cell-center evaluation or integrated cell mass, how rotations are interpolated, and whether truncation is renormalized. Report the retained mass percentile and its sensitivity.

5.4 Environment-aware transition matrices

This section is anchored in Sections 3.4–3.5 of the coauthored random-walk manuscript (Buchin, Hagedoorn, and Chatila 2026). Let $G \subset \mathbb{Z}^2$ be the grid domain and let $\ell : G \to \mathcal{L}$ map each cell to a terrain class. A subset $\mathcal{B} \subseteq \mathcal{L}$ contains barrier classes and $\mathcal{P} = \mathcal{L} \setminus \mathcal{B}$ contains passable classes. The base matrix $M_{x,y,d,t}$ is post-processed before use by the dynamic program, so environmental constraints can depend on location, incoming direction, and time without changing the general recurrence.

5.4.1 Hard reachability for barriers

Checking only the destination cell would allow a long transition to jump across a narrow barrier. Hard reachability therefore tests every cell intersected by the rasterized segment from source p to destination $p + \delta$. With $\mathrm{ray}(p, p+\delta)$ computed by Bresenham's line algorithm (Bresenham 1965), define

$$H_{\mathrm{hard}}(p, \delta) = \begin{cases} 1, & \ell(q) \notin \mathcal{B} \text{ for all } q \in \mathrm{ray}(p, p+\delta), \\ 0, & \text{otherwise.} \end{cases} \tag{5.4}$$

If the source lies on a barrier, the hard-reachability matrix is zero. If the entire local neighborhood is barrier-free, the base matrix is reused. Otherwise,

$$M^{H}_{x,y,d,t} = \mathrm{normalize}\big(M_{x,y,d,t} \odot H_{\mathrm{hard}}(p, \cdot)\big) . \tag{5.5}$$

A zero sum after masking indicates that no local transition is feasible and must remain distinguishable from an ordinary normalized kernel.

5.4.2 Relaxed barrier handling

Relaxed handling covers two cases: a passable source whose route to the target crosses discouraged terrain, and a source already located on barrier terrain. For the first case, let

$$\pi(p, p+\delta) = (q_0, \ldots, q_m), \qquad q_0 = p, \quad q_m = p+\delta, \tag{5.6}$$

be the rasterized path. A multiplicative path factor accumulates terrain-pair weights along the path,

$$\rho(p,\delta) = \prod_{i=1}^{m} \tau\big(\ell(q_{i-1}), \ell(q_i)\big), \tag{5.7}$$

where $\tau : \mathcal{L} \times \mathcal{L} \to \mathbb{R}_{\geq 0}$. A separate endpoint factor $\eta(p,\delta)$ may penalize transitions that finish inside a barrier. The adjusted matrix is

$$M^r_{x,y,d,t} = \text{normalize}\Big(\big(M_{x,y,d,t}(\delta)\,\rho(p,\delta)\,\eta(p,\delta)\big)_{\delta\in[-R,R]^2}\Big). \tag{5.8}$$

When the source itself lies inside a barrier, the manuscript uses a directional shore-seeking bias. Rays from the transition-window boundary estimate the nearest passable cell in each direction bin; if c_d is the shortest finite shore distance for bin d, then

$$w_d = \begin{cases} \left(1 - \dfrac{c_d}{\sum_{j:c_j<\infty} c_j}\right)^{\gamma}, & c_d < \infty, \\ 0, & c_d = \infty, \end{cases} \qquad \gamma > 1. \tag{5.9}$$

The directional matrix is scaled by w_d without normalizing each direction separately, because such a normalization would remove the intended directional preference.

Writing prompt

Preserve the paper's two-case distinction when expanding this section. Add implementation edge cases: grid boundary, no visible shore, source/end on barriers, repeated cells in a rasterized ray, and the semantics of a zero transition matrix.

5.4.3 General terrain-transition weights

Barrier penalties are a special case of a general matrix $W \in \mathbb{R}_{\geq 0}^{|\mathcal{L}|\times|\mathcal{L}|}$, where W_{ab} weights a consecutive transition from class a to class b. These values are multiplicative path factors and therefore need not be row-stochastic; the resulting displacement matrix is normalized only after all factors are applied.

Explain whether W is expert-specified, learned from used/available terrain transitions, or estimated jointly with another model. Report identifiability constraints, smoothing for unseen pairs, uncertainty, and sensitivity.

Table 5.1: Template for an estimated or specified terrain-transition weight matrix. Diagonal values may be fixed to one for identifiability.

From / to	Forest	Sand	Grass	Water
Forest	1.00	—	—	—
Sand	—	1.00	—	—
Grass	—	—	1.00	—
Water	—	—	—	1.00

5.5 Combining baseline and environmental parameters

When dense environmental fields are mapped to movement parameters, they may be combined with parameters estimated from tracking data. For a component-wise weight vector $\mathbf{w} \in [0,1]^p$,

$$\boldsymbol{\eta}(x,y,t) = (\mathbf{1} - \mathbf{w}) \odot \boldsymbol{\eta}_{\text{base}} + \mathbf{w} \odot \boldsymbol{\eta}_{\text{env}}(x,y,t). \tag{5.10}$$

This parameter interpolation is meaningful only when both vectors share definitions, units, scale, and admissible ranges.

5.6 Kernel diagnostics

Kernel validation includes normalization, symmetry where expected, mass-retention checks, reachability, stability across grid resolutions, and comparison between simulated and held-out distributions of step length and turning angle. State-specific diagnostics also report training counts and uncertainty.

Add a multi-panel figure showing representative Brownian, correlated, state-dependent, and terrain-adjusted kernels on a common physical scale.

6 Endpoint-Conditioned Random-Walk Interpolation

6.1 Discrete model

Space is represented by a finite regular grid and time by $t \in \{0, \dots, T\}$. A path $X_{0:T}$ starts at cell $\mathbf{s}$ and is conditioned to end at cell $\mathbf{e}$. For a first-order walk, a transition from cell u to v at time t has probability $P_t(v \mid u)$. Directionally correlated walks extend the state by an incoming heading bin d.

The probability of a particular conditioned path is

$$\Pr(X_{0:T} = x_{0:T} \mid X_0 = \mathbf{s}, X_T = \mathbf{e}) = \frac{\prod_{t=0}^{T-1} P_t(x_{t+1} \mid x_t)}{\Pr(X_T = \mathbf{e} \mid X_0 = \mathbf{s})}, \tag{6.1}$$

for $x_0 = \mathbf{s}$ and $x_T = \mathbf{e}$.

6.2 Dynamic programming

The forward recurrence accumulates the probability of reaching each state:

$$F_{t+1}(v) = \sum_{u \in \mathcal{N}^-(v)} F_t(u) P_t(v \mid u), \quad F_0(u) = \mathbb{1}[u = \mathbf{s}]. \tag{6.2}$$

The direction-aware version performs the analogous sum over predecessor cells and heading bins. The endpoint mass $F_T(\mathbf{e})$ is also the normalizing constant in Equation (6.1). A value of zero means the chosen space–time discretization cannot connect the observations under the model.

6.3 Backward path sampling

Starting at $X_T = \mathbf{e}$, predecessor states can be sampled with probability proportional to

$$\Pr(X_t = u \mid X_{t+1} = v, X_0 = \mathbf{s}) \propto F_t(u) P_t(v \mid u). \tag{6.3}$$

Repeating this operation yields exact samples from the discretized endpoint-conditioned model, up to numerical error and the finite computational domain.

6.4 Step-count and grid policy

The relationship among the observed elapsed time, kernel interval, grid resolution, support radius, and number of transitions must be explicit. If a kernel represents physical duration Δt_{model}, an initial choice is

$$T = \text{round}\left(\frac{t_{i+1} - t_i}{\Delta t_{\text{model}}}\right). \tag{6.4}$$

Reachability requires TS to be at least the endpoint distance in grid steps under the available transitions. If this fails, increasing T, enlarging support, or rejecting the interval corresponds to different model assumptions and must be reported.

6.5 Boundaries, barriers, and context-dependent transitions

Transitions leaving the finite computational window or entering hard barriers receive zero mass. Renormalizing the remaining transitions conditions the local step on remaining within the modeled domain; discarding mass without local renormalization has another interpretation. The selected convention should be consistent between forward computation, backward sampling, and spatial summaries.

Writing prompt

Specify the exact semantics of domain boundaries, barrier cells, start/end cells located on barriers, and no-path failures. Explain any relaxed-barrier policy separately.

6.6 Numerical stability and implementation

Long sequences can underflow when probabilities are multiplied repeatedly. Possible remedies include per-layer normalization with retained scale factors, log-domain computations, or higher precision. Parallel central processing unit (CPU) and graphics processing unit (GPU) implementations must preserve normalization and sampling semantics, even if floating-point reduction order leads to small differences.

6.7 Output and uncertainty communication

A collection of sampled paths is a Monte Carlo representation, not a set of repeated observations. Visualizations should show endpoints, environmental context, and uncertainty without suggesting that individual samples are equally likely ground truth. Where possible, marginal probability surfaces or credible regions should accompany a small number of sampled paths.

Add pseudocode for forward computation and backward sampling, with symbols aligned to the implementation. Include complexity in terms of grid cells, time steps, direction bins, and kernel support.

7 Utilization Distributions and Visit Probabilities

7.1 Conditioned marginal occupancy

For one observation interval, define a backward message

$$B_t(u) = \Pr(X_T = \mathbf{e} \mid X_t = u). \tag{7.1}$$

Combining forward and backward messages gives the conditioned marginal probability of occupying cell u at time t:

$$\gamma_t(u) = \Pr(X_t = u \mid X_0 = \mathbf{s}, X_T = \mathbf{e}) = \frac{F_t(u)B_t(u)}{F_T(\mathbf{e})}. \tag{7.2}$$

The finite-interval occupancy distribution is obtained by aggregating γ_t with explicit temporal weights:

$$U(u) = \frac{\sum_{t=0}^{T} \omega_t \gamma_t(u)}{\sum_{t=0}^{T} \omega_t}. \tag{7.3}$$

Uniform weights estimate the distribution at a uniformly selected model time. Alternative weights may be required when model intervals represent unequal duration.

7.2 Combining observation intervals

Track-level spatial use combines interval-level distributions. Weighting each interval equally differs from weighting it by duration, transition count, or confidence. Endpoints may also be double-counted when adjacent intervals are added independently. The aggregation rule therefore forms part of the estimand, not merely the implementation.

Writing prompt

State whether the final UD is conditional on the observed fixes, how observation times contribute, and how multiple animals are pooled. Reserve “home range” for an estimand and time horizon that justify the term.

7.3 Isopleths and raster output

Probability-mass contours are derived by sorting cells by density (or mass, for equal-area cells) and accumulating until a target level is reached. Common labels such as "50% core area" and "95% range" should specify whether values represent cell mass or a density threshold. Geographic export requires correct pixel alignment, affine transform, CRS, no-data semantics, and units.

7.4 Visit probability

For target region A, direct computation of the probability of ever visiting A can use a complementary DP. Let $F_t^{\neg A}(u)$ propagate only paths that have not entered A. Then

$$P_{\text{visit}}(A) = 1 - \frac{F_T^{\neg A}(\mathbf{e})}{F_T(\mathbf{e})}, \tag{7.4}$$

provided the same conditioning and boundary conventions are used in numerator and denominator. This quantity generally cannot be recovered by summing marginal occupancies because a path may occupy the target on multiple time steps.

7.5 Monte Carlo and exact estimators

Spatial summaries can be computed exactly on the discretized state space or estimated from sampled paths. Exact DP avoids Monte Carlo variance but may require additional memory or passes. Sampling supports richer path functionals but should report uncertainty as a function of the number of walks. Cross-checking exact and sampled estimates is a valuable implementation test.

7.6 Parallel and GPU computation

The spatial-summary algorithms are a thesis extension beyond the reference manuscript. On the CPU, the backward utilization update scatters contributions from a destination state to many predecessor states. A direct OpenMP implementation therefore needs synchronization. The implementation provides an atomic-update variant and a thread-local variant that accumulates private state tensors before a parallel reduction. The former minimizes auxiliary memory; the latter avoids frequent atomics at the cost of memory proportional to the number of worker threads.

The GPU implementation instead assigns work to predecessor states and gathers contributions from reachable destinations. It maintains the current and previous utilization layers, transition denominators, a packed kernel pool, per-cell kernel indices, and direction

metadata on the device. Forward-density layers are streamed as needed. Two output policies are relevant:

Full time-resolved distribution
Copy or retain every conditioned layer when time-specific output is required.

Memory-bounded aggregate
Accumulate directions and time directly into one two-dimensional matrix, retaining only two utilization layers and the final sum.

Writing prompt

Present these algorithms as new thesis contributions only to the extent supported by the final contribution statement. Give pseudocode, work and memory complexity, data-transfer costs, and equivalence tests against the serial reference. Distinguish kernel execution time from end-to-end time including host/device transfer.

7.7 Comparison with reference methods

Reference comparisons should align coordinate system, extent, cell size, normalization, contour convention, time window, and included fixes. Candidate baselines include empirical held-out occupancy, bridge-based methods, conventional kernel density estimate (KDE), and autocorrelated UDs. Agreement may be assessed through integrated absolute error, overlap coefficients, divergence measures, contour area, calibration, and likelihood on held-out locations.

Insert the final definition of every reported metric and explain its direction, scale, and failure modes. Predefine which metric answers which research question.

8 Software Architecture and MoveApps Integration

8.1 Design goals

The implementation aims for modularity, explicit configuration, preservation of spatial metadata, informative failures, and reproducible artifacts. Each pipeline stage accepts and returns a documented data contract so that it can be tested independently and composed in alternative workflows.

8.2 Component architecture

8.3 Data and metadata contract

Core fields include the individual identifier, timestamp, geometry, source CRS, projected coordinates, and any terrain or state annotation. Metadata propagated between components should record units, model interval, grid origin and cell size, class/state mappings, parameter configuration, software version, and warnings. A valid output is complete and internally consistent; partial annotation should not be returned silently after an error.

8.4 Python and native implementation boundary

The Python layer provides data ingestion, geospatial processing, model configuration, diagnostics, and workflow integration. Computationally intensive random-walk operations are implemented in a native library, with optional parallel CPU and GPU back ends. The binding layer owns memory-layout conversion, error propagation, parameter validation, and deterministic seed handling.

Writing prompt

Add a diagram of Python packages, native library, external data services, and MoveApps boundaries. Document ownership and lifetime of arrays crossing the language boundary, especially for GPU execution.

Table 8.1: Proposed component responsibilities and principal contracts.

Component	Input	Output
Environmental annotation	Trajectory collection with valid CRS	Terrain/context columns, projected coordinates, optional raster artifact
Trajectory segmentation	Ordered fixes and gap/criterion configuration	Segment identifiers and quality flags
Behavioral annotation	Valid movement features per segment	States, change points, diagnostics, model metadata
Kernel construction	Annotated movement steps	Normalized kernels with physical interval and support metadata
Random-walk interpolation	Endpoints, kernels, grid/context, policy	Sampled conditioned paths and diagnostics
Spatial summaries	Conditioned interval model	UD, contours, visit probabilities, raster/vector artifacts

8.5 Transition-matrix construction modes

The native random-walk implementation supports three strategies inherited from the manuscript's system design (Buchin, Hagedoorn, and Chatila 2026):

On-the-fly construction
: Recompute the environment-adjusted matrix whenever it is needed. This has the smallest persistent memory footprint but repeats work.

Hash-table precomputation
: Cache each distinct combination of terrain kernel and local reachability mask; grid cells reference the cached matrix. Performance depends on the diversity of local terrain configurations.

Serialization
: Persist transition maps or dynamic-programming layers and stream them when required, exchanging compute or I/O time for lower peak memory.

8.6 Parallel spatial-summary back ends

The thesis implementation extends the native library with multiple UD execution strategies. The serial implementation provides a correctness reference. Two OpenMP variants use either atomic scatter updates or thread-local accumulation followed by reduction. The CUDA mixed-walker implementation packs variable terrain kernels into contiguous buffers, constructs device-side direction metadata, computes normalization denominators,

and propagates the conditioned utilization distribution backward with a gather kernel. A reduced CUDA interface produces the time-averaged two-dimensional UD without materializing the complete time series.

Document the final public API names and supported combinations of Brownian, correlated, state-dependent, and mixed kernels. Add a feature matrix for serial CPU, OpenMP atomic, OpenMP thread-local, and CUDA implementations.

Writing prompt

The coauthored paper already contains CPU/GPU work for the baseline dynamic program. Clearly mark GPU UD computation and the parallel UD variants as subsequent thesis work, and identify your implementation and experimental contributions precisely.

8.7 Configuration and provenance

Every run should serialize a machine-readable configuration containing input identifiers, component versions, parameter values, random seeds, and output locations. External environmental queries additionally record product and asset identifiers. Repository commit hashes are preferable to branch names because working branches and uncommitted changes are not stable provenance.

8.8 Testing strategy

The test portfolio should include:

- unit tests for coordinates, normalization, feature calculation, and parameter validation;
- property tests for probability conservation and conditioned endpoints;
- synthetic cases with analytically known or exhaustively enumerable results;
- integration tests spanning the Python/native and component boundaries;
- regression tests for representative terrestrial, marine, and wide-ranging data;
- numerical comparisons between serial, parallel, and GPU back ends.

8.9 User-facing workflow

Present one concise end-to-end workflow with screenshots or a configuration table. Explain the ecological meaning of exposed settings; do not mirror internal implementation parameters that users cannot choose responsibly.

8.10 Software limitations

The software boundary does not remove modeling assumptions. Raster availability, projection distortion, data volume, insufficient observations for rare states, unreachable endpoint pairs, and hardware-specific resource limits should produce actionable diagnostics. Service availability for external environmental products is also distinct from algorithmic correctness.

9 Evaluation Design

9.1 Evaluation principles

The evaluation separates scientific validity, numerical correctness, robustness, and computational performance. Parameters selected on one data subset should be assessed on held-out intervals or case studies wherever possible. The evaluation unit (fix, interval, trajectory, individual, or study) must be declared for every metric to avoid pseudo-replication.

9.2 Data splits and missingness experiments

A useful interpolation experiment treats observed high-frequency fixes as partial ground truth. Interior fixes are hidden according to controlled sampling schemes, the retained endpoints condition the model, and predictions are evaluated against the withheld locations. Schemes should include regular thinning, randomly missing fixes, and contiguous gaps of different duration.

Writing prompt

Fix all data splits and seeds before final comparison. Prevent leakage: hidden fixes must not be used to fit state labels, movement kernels, scaling parameters, or tuning choices for the evaluated interval unless the experiment explicitly studies that setting.

9.3 Baselines and ablations

Candidate baselines include linear interpolation, Brownian or dynamic Brownian bridges, and relevant Gaussian-process or autocorrelated movement models. Ablations isolate the value of each pipeline component:

- environment-aware versus environment-agnostic transitions;
- state-dependent versus pooled kernels;
- Brownian versus correlated kernels;
- fixed versus data-adaptive support;
- exact summaries versus sampled-walk estimates;
- segmentation/annotation methods under a common downstream model.

9.4 Predictive and spatial metrics

For held-out locations, report proper scoring rules or log probability when available, distance to high-probability regions, coverage of nominal mass contours, and spatial overlap with an empirical reference. For UDs, possible metrics include integrated absolute error,

$$\mathrm{IAE}(P, Q) = \frac{1}{2} \sum_{u \in \mathcal{G}} |P(u) - Q(u)|, \tag{9.1}$$

and the Bhattacharyya coefficient,

$$\mathrm{BC}(P, Q) = \sum_{u \in \mathcal{G}} \sqrt{P(u)Q(u)}. \tag{9.2}$$

Both require aligned, normalized grids.

9.5 Robustness and sensitivity

Sensitivity analyses vary grid resolution, kernel mass percentile, number of direction bins, state count, barrier weight, model interval, number of sampled walks, and gap duration. Results should distinguish gradual sensitivity from qualitative failure, such as loss of reachability or a state with insufficient training steps.

9.6 Computational benchmarks

Benchmark the complete workflow and major stages separately. For the random-walk core, vary grid dimensions, T, kernel support, number of direction bins, and requested samples. Compare serial and parallel CPU implementations and, where available, GPU acceleration. Report wall-clock time, memory, hardware, compiler/build flags, warm-up policy, number of repetitions, and a robust summary such as median with interquartile range.

9.6.1 Spatial-summary back ends

Evaluate serial, OpenMP atomic, OpenMP thread-local, and CUDA UD calculations on identical inputs. Compare both the full time-resolved output and the memory-bounded time aggregate. The validation must precede speed comparisons: record maximum absolute and relative error, normalization error per time layer, aggregate-grid error, and behavior for unreachable endpoints. Performance measurements should separately report setup, host/device transfer, kernel execution, reduction, and total wall time.

Table 9.1: Evaluation matrix to complete before running final experiments.

Claim	Comparison	Primary metric	Evaluation unit
Better gap interpolation	Proposed model vs baselines	Held-out log score / coverage	Interval within individual
Environmental realism	Context model vs ablation	Barrier violations / score	Interval or path
Improved spatial summary	Proposed UD vs references	IAE / BC / contour coverage	Individual
Scalable computation	CPU/GPU variants	Runtime and peak memory	Configuration

9.6.2 End-to-end application pipeline

Measure environmental acquisition/annotation, segmentation, behavioral fitting, kernel estimation, random-walk preprocessing, interpolation, and spatial summaries separately. This exposes the actual bottleneck in realistic MoveApps runs rather than extrapolating from the random-walk core alone. Include at least one local detailed-land-cover study and one long-range land/water study if these modes remain in the final software.

9.7 Statistical reporting

Aggregate results across individuals rather than only across fixes, display the distribution of paired differences, and report uncertainty intervals. If inferential models are used, describe dependence structure and justify the sampling unit. Exploratory comparisons should be labeled as such; a large number of metrics should not be converted into selective claims.

9.8 Reproducibility plan

Freeze an experiment manifest linking each table and figure to a command, configuration file, software revision, and raw output. Record exclusions before examining final metric values.

10 Results

Writing prompt

This chapter is deliberately a results shell. Report observations and estimates here; reserve explanations, implications, and speculation for Chapter 11. Keep the subsection order aligned with the research questions and evaluation plan.

10.1 Data and preprocessing outcomes

Report sample sizes, fix intervals, exclusions, projection modes, environmental coverage, annotation failures, and the number and duration of resulting segments. Include a flow diagram or accounting table so all observations are traceable.

10.2 Behavioral annotation and kernel estimates

Report model-selection diagnostics, state occupancy, feature summaries, transition behavior, stability, and kernel parameters/support. Show representative tracks and kernels without assigning ecological labels solely from numeric state order.

10.3 Interpolation quality

For each baseline and ablation, report the preregistered primary metrics on held-out intervals. Include paired distributions across individuals, coverage by gap duration, and representative success and failure cases.

10.4 Utilization distributions and visit probabilities

Report aligned-grid comparisons, contour areas and coverage, exact-versus-sampled agreement, and visit-probability calibration where event labels are available. State the temporal interpretation of every surface.

Table 10.1: Template for held-out interpolation results. Values should include an across-individual uncertainty summary.

Method	Log score ↑	95% coverage → 0.95	Distance ↓	Invalid intervals ↓
Linear interpolation	—	—	—	—
Brownian bridge	—	—	—	—
Pooled random walk	—	—	—	—
State-aware random walk	—	—	—	—
Environment-aware model	—	—	—	—

Table 10.2: Research-question synthesis to complete after the analysis.

Question	Main evidence	Answer / qualification
RQ1	—	—
RQ2	—	—
RQ3	—	—
RQ4	—	—
RQ5	—	—

10.5 Sensitivity and failure analysis

Summarize sensitivity to grid size, support percentile, state count, direction bins, time policy, barrier weights, and sample count. Quantify no-path cases, numerical issues, and resource exhaustion rather than omitting them.

10.6 Computational performance

Report runtime and memory scaling with hardware/software details. Separate one-off preprocessing and kernel fitting from per-interval dynamic programming and per-path sampling. Give a dedicated comparison of serial, OpenMP atomic, OpenMP thread-local, and CUDA UD calculations, including numerical equivalence and full-series versus memory-bounded aggregation. Include CPU/GPU crossover points only when supported by end-to-end measurements.

10.7 Summary of findings

11 Discussion

11.1 Principal findings

Answer each research question in the same order as Section 1.3. Lead with the magnitude and uncertainty of the relevant result, then interpret it. Distinguish what the data show from what is inferred.

11.2 Ecological interpretation

Environment- and state-dependent paths can be more plausible without necessarily identifying the animal's actual route. Interpret improvements in probabilistic prediction as better representation of uncertainty under the chosen model. Discuss whether the derived space-use surfaces answer questions about finite observation gaps, recurring habitat use, or longer-term home range.

Relate the findings to the ecology of the selected species and scale. Address how classification error, fix bias, seasonality, individual variation, and environmental-product resolution constrain interpretation.

11.3 Methodological implications

Discuss what is gained by arbitrary transition kernels relative to Brownian bridges and what costs arise from discretization, parameter estimation, and computational complexity. Identify situations where the simpler baseline remains preferable.

11.4 Software and workflow implications

Modular components make preprocessing decisions inspectable and reusable, but they also expose interfaces at which units or metadata can be lost. Discuss which guarantees are enforced by the implementation, which rely on user configuration, and how MoveApps packaging changes reproducibility and accessibility.

11.5 Limitations and threats to validity

At minimum, consider the following limitations:

- observed locations are noisy and missingness may be non-random;
- latent states are model-dependent and not direct behavioral observations;
- environmental layers differ in resolution, accuracy, and reference time;
- discretization introduces spatial, temporal, directional, and boundary effects;
- rare states or terrain transitions may be weakly estimated;
- case studies may not generalize across species, landscapes, or fix schedules;
- reference UDs are comparison methods, not error-free ground truth;
- GPU and parallel results depend on hardware and implementation details.

Writing prompt

For each important limitation, state its likely direction of bias, which results it affects, and any diagnostic or mitigation. Avoid a disconnected list of generic caveats.

11.6 Future work

Potential extensions include uncertainty-aware environmental annotation, continuous-time or adaptive grids, hierarchical kernels across individuals, joint inference of behavior and movement, dynamic weather or current fields, calibrated terrain-transition learning, and more memory-efficient exact spatial summaries.

Prioritize future work by scientific value and feasibility. Separate immediate engineering work from new research questions.

12 Conclusion

Sparse tracking observations leave uncertainty precisely where many ecological questions concern movement and exposure. This thesis frames that problem as an end-to-end pipeline: environmental context and trajectory structure inform movement kernels; endpoint-conditioned random walks represent plausible paths; and dynamic programming supports interpolation, utilization distributions, and visit probabilities within one model family.

Rewrite this chapter after the results and discussion. In two to four pages: restate the problem, give direct answers to the research questions with the key quantitative evidence, identify the main practical contribution, and close with the broader ecological significance. Introduce no new results or citations.

A Reproducibility Record

A.1 Software revisions

Record immutable revisions used for the submitted results.

A.2 Computing environment

Record operating system, Python and R versions, environment lock files, compiler and CUDA versions, relevant build flags, CPU, GPU, RAM, and storage. Include the exact commands used for final figures and tables.

A.3 Data provenance

For each data set, record DOI or stable identifier, download date, licence, access restrictions, original CRS/time zone, checksums, and all transformations. Do not archive restricted animal locations in a public supplement.

Table A.1: Software provenance.

Component	Repository / package	Commit or version
Environmental annotation	`cma_landcover`	—
Criteria segmentation	`segmentationcma`	—
Behavioral annotation	`hmmcma`	—
Random walks / Python interface	`RW-Python-gitlab`	—
Native random-walk library	nested native project	—
Thesis sources	this repository	—

Table A.2: Traceability manifest for final outputs.

Thesis output	Command / script	Configuration	Raw result artifact
Table 10.1	—	—	—
Table 10.2	—	—	—
Primary UD figure	—	—	—
Runtime figure	—	—	—

A.4 Experiment manifest

A.5 Randomness and determinism

List the seed hierarchy for data splitting, model initialization, mixture fitting, random-walk sampling, and bootstrap intervals. For parallel or GPU routines that are not bitwise deterministic, document expected numerical variability and the tolerance used in regression tests.

B Supplementary Definitions and Results

B.1 Notation

B.2 Extended diagnostics

Place supporting sensitivity plots, per-individual results, additional kernel visualizations, and diagnostic tables here. Every appendix item should be referenced from the main text.

B.3 Configuration schemas

Include final user-facing configuration tables or concise schemas when they are necessary to reproduce the work. Avoid long source-code listings; link to versioned software instead.

Table B.1: Core notation used throughout the thesis.

Symbol	Meaning
$\mathcal{T}$	Observed time-stamped trajectory
$\mathbf{s}, \mathbf{e}$	Start and end cells of one conditioned interval
X_t	Random-walk state at discrete time t
T	Number of transitions within an observation interval
c	Spatial grid-cell width
R, S	Physical and discrete kernel-support radii
$P_t(v \mid u)$	Transition probability from state u to v
F_t, B_t	Forward and backward dynamic-programming messages
$\gamma_t(u)$	Endpoint-conditioned marginal occupancy
$U(u)$	Time-aggregated finite-interval occupancy distribution

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